

Lecture 1.10 – Troubleshooting Problems

Specific Learning Objectives:

3.5.1 – Learn basic skills in debugging and troubleshooting error messages.

What happens when you get an error?



- Errors are quite normal. No one is perfect (far from it!), and mistakes slip through all the time.
- Learning a language (spoken or technical) involves making a lot of errors! You have to be comfortable being bad at something in order to practice and learn it.
- Troubleshooting errors can be challenging and frustrating! But there is a solution.
- Today, we'll cover some strategies for tackling errors.



Strategies for Troubleshooting Code



- First, take a deep breath and remember you can fix it. And if you can't figure it out, someone can help you.

- Follow the simple steps below:
 1. Did you spell it correctly? *Are you sure?*
 2. Is the capitalization/punctuation correct?
 3. Is the syntax correct?
 4. Is the object in your environment?

1. Did you spell it correctly?

- About 75% of students who come for help with an error in the first half of this course have spelled something incorrectly.
- Remember that object and function names must be *spelled exactly correct*. There is no autocorrect, you have to get it right! (But there is actually autofill, which helps!)
- What do errors look like when you spell things incorrectly?
Example: Loblolly data set.

```
> View(Loblolly)
Error in View : object 'Loblolly' not found
> View(Lobolly)
Error in View : object 'Lobolly' not found
> View(Bloblolly)
Error in View : object 'Bloblolly' not found
```

Please check your spelling before panicking about errors!



R is looking for *exactly* what you tell it to look for!

2. Is the capitalization/punctuation correct?

- If things are spelled correctly, make sure the punctuation and capitalization is correct.
- Remember, *R* is case-sensitive (like many passwords): Loblolly is different than loblolly.
- Similarly, punctuation has to be correct as well. `dat.model` is not the same as `dat_model`.
- Errors will look the same as misspellings: *R* will tell you it can't find an object!

me: *gets mad at code for not doing what I coded it to do*

the code doing exactly what I coded it to do:



Please check your capitalization before panicking about errors!

3. Is the syntax correct?

- Syntax errors are also common for both beginning and experienced programmers! (They can be harder for beginners to find.)
- Remember, syntax is incredibly important and the most difficult part of learning a language.
- What will errors look like? They range from very straightforward to very unclear.

Unmatched square bracket

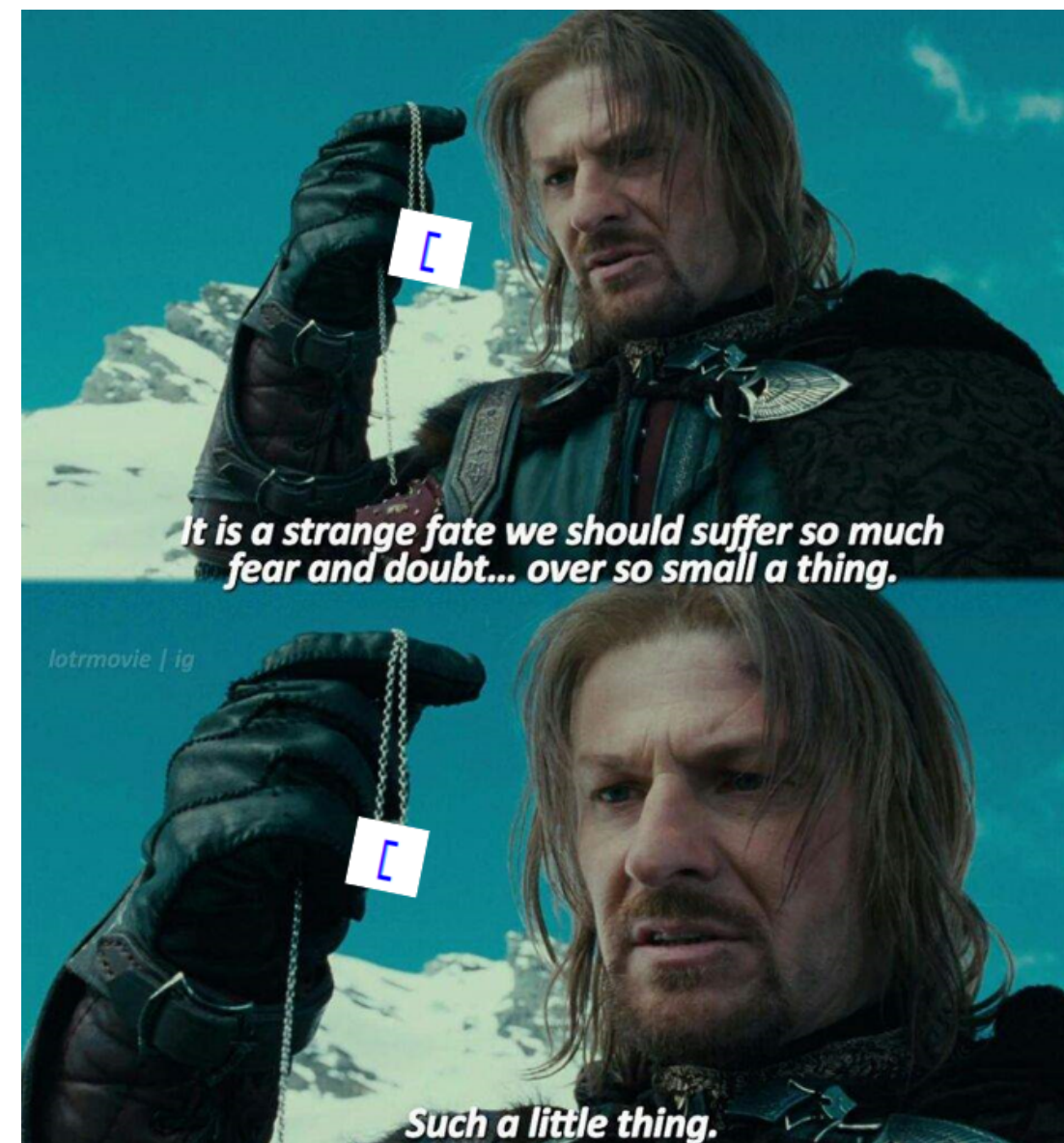
```
> d[ <- 3
```

Error: unexpected assignment in "d[<-"

Unmatched parenthesis

```
> d <- c(3, 5, 5
```

(R is waiting for you to close this!!)



Can't use minus sign for object name!

```
> pop-v <- 4
```

Error in pop - v <- 4 : object 'pop' not found

Can't start an object name with a number!

```
> 3flips <- 3
```

Error: unexpected symbol in "3flips"

Syntax tips

- Object naming rules:

- Object names must begin with a letter.

Correct: `flip3`, `flip.3`, `Flip_3`

Incorrect: `3flip`, `_flip3`, `=flip3`

- Object names must not contain special characters or spaces (stick with `.` and `_`).

Correct: `flip3`, `flip.3`, `Flip_3`

Incorrect: `flip-3`, `flip#3`, `Flips@3`

- Avoid renaming already existing objects

Correct: `dat`, `t`, `name1`

Incorrect: `data`, `T`, `names`

Syntax tips

- Avoid problems with brackets and parentheses by using whitespace! Making code more human-readable will help you find errors faster.
- R is not sensitive to whitespace, so use spaces and tabs!

#Bad

```
df<-data.frame(x=c(9,2,54,1,39,99,29,40,80,2,68,3,34),y=c(T,F,T,T,T,F,T,F,F,T,F,F,T))
```

Is the y value that corresponds to x = 39 True or False?

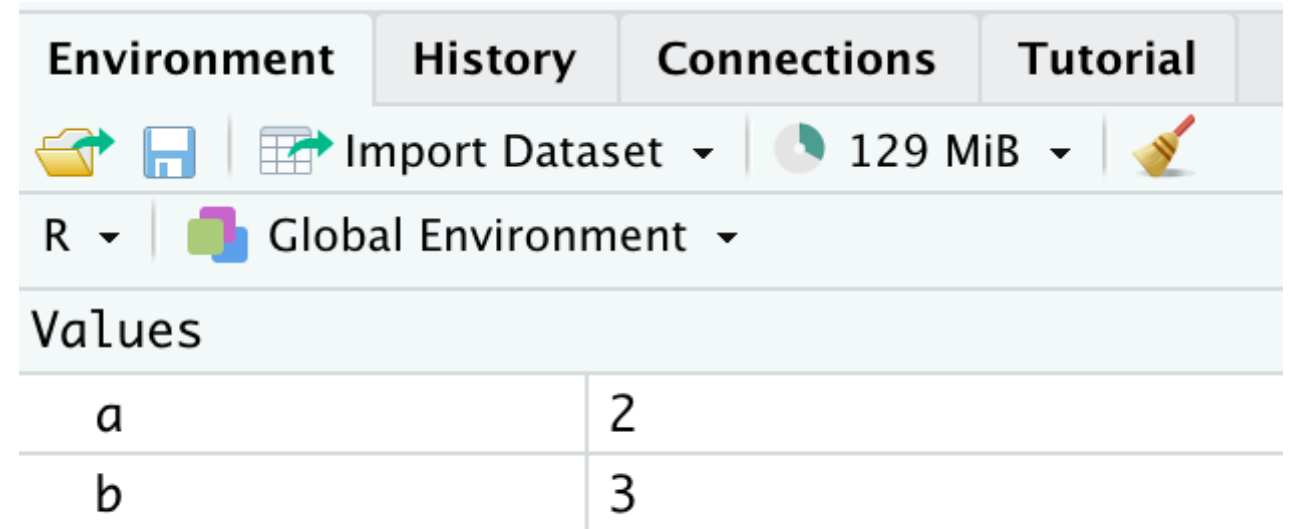
Good

```
df <- data.frame(x = c(9, 2, 54, 1, 39, 99, 29, 40, 80, 2, 68, 3),  
                 y = c(T, F, T, T, T, F, T, F, F, T, F, F))
```

TRUE!

4. Is the object in your environment?

- The environment lists all objects in R's memory.
- It will be empty every time R starts or restarts.
- Sometimes, you'll assume objects are in the environment and they won't be there (or they won't be the same). This will result in an error where there was not before!
- Tips for avoiding this:
 - Write all your code in scripts, in the order in which each line should be run.
 - Check the environment in RStudio to make sure it's there.
 - Frequently restart R or clear your environment to make sure your script runs cleanly!



The screenshot shows the RStudio interface with the 'Environment' tab selected. It displays the 'Global Environment' containing two objects: 'a' with value 2 and 'b' with value 3. The 'Values' section is visible below the environment list.

Values	
a	2
b	3

Troubleshooting Strategy: Splitting

- What do you do if you don't know which part of the code is the problem?
- Try splitting the line into the smallest elements. Run each independently to try and pinpoint the problem.
- Example: many ways to split the statement.



The diagram illustrates the process of splitting the R code line into segments for troubleshooting. The code line is: `> mean(Indometh$conc[Indometh$Time==8.00])`. Above the code, several horizontal lines of different colors (red, orange, green, teal, purple, blue) are positioned to show how the code can be split into smaller, testable parts. For example, the first red line is above the opening parenthesis, the second red line is above the closing parenthesis, the orange line is above the entire expression in brackets, the green line is above the variable `Indometh$conc`, and the teal line is above the indexing part `[Indometh$Time==8.00]`. The purple line is above the entire expression `Indometh$conc[Indometh$Time==8.00]`, and the blue line is above the `mean()` function call.

```
> mean(Indometh$conc[Indometh$Time==8.00])  
[1] NaN
```

Troubleshooting Strategy: Splitting

```
> mean(Indometh$conc[Indometh$Time==8.00])
```

```
[1] NaN
```

```
2 mean(Indometh$conc[Indometh$Time==8.00])
```

```
3
```

```
4
```

```
> Indometh$conc
```

```
[1] 1.50 0.94 0.78 0.48 0.37 0.19 0.12 0.11 0.08 0.07 0.05 2.03 1.63 0.71 0.70 0.64  
[17] 0.36 0.32 0.20 0.25 0.12 0.08 2.72 1.49 1.16 0.80 0.80 0.39 0.22 0.12 0.11 0.08  
[33] 0.08 1.85 1.39 1.02 0.89 0.59 0.40 0.16 0.11 0.10 0.07 0.07 2.05 1.04 0.81 0.39  
[49] 0.30 0.23 0.13 0.11 0.08 0.10 0.06 2.31 1.44 1.03 0.84 0.64 0.42 0.24 0.17 0.13  
[65] 0.10 0.09
```

```
1  
2 mean(Indometh$conc[Indometh$Time==8.00])
```

```
3
```

```
4
```

```
1  
2 mean(Indometh$conc[Indometh$Time==8.00])
```

```
3
```

```
4
```

```
> Indometh$Time==8.00
```

```
logical(0)
```

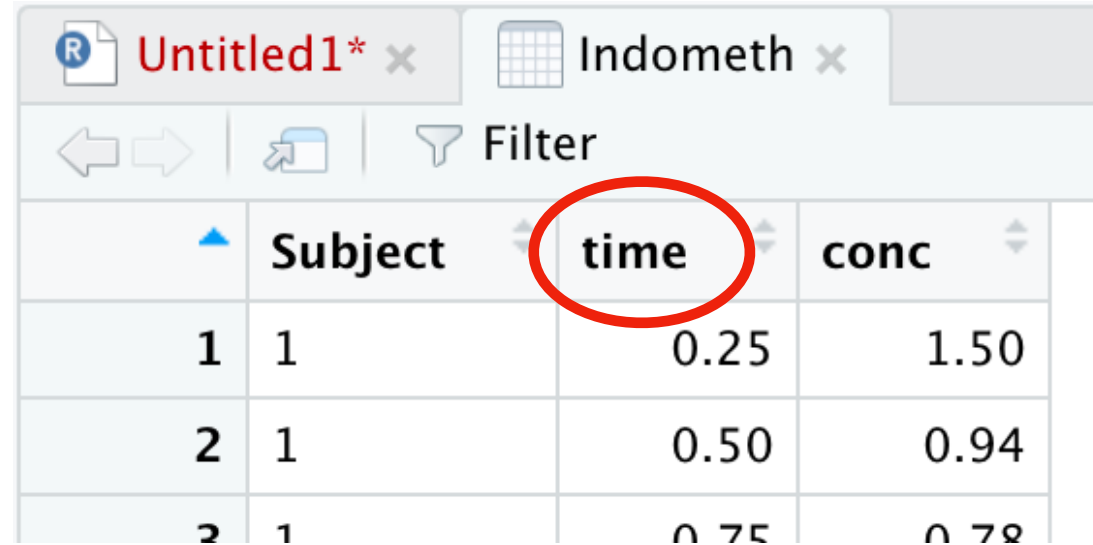
```
> Indometh$Time
```

```
NULL
```

Troubleshooting Strategy: Splitting

```
> mean(Indometh$conc[Indometh$Time==8.00])  
[1] NaN
```

```
> View(Indometh)
```



	Subject	time	conc
1	1	0.25	1.50
2	1	0.50	0.94
3	1	0.75	0.78

Fix it:

```
> mean(Indometh$conc[Indometh$time==8.00])  
[1] 0.07166667
```

If you are stuck, try taking a break!

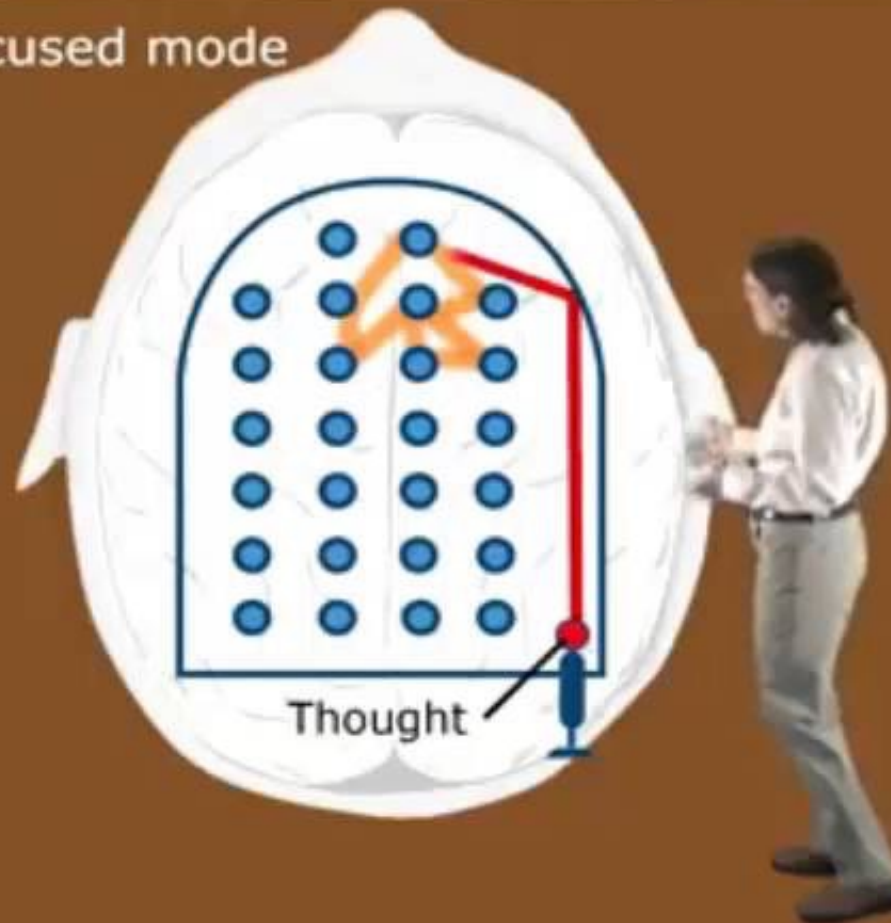
Introduction to the Focused and Diffuse Modes



Learning
How to Learn

Bernhard Egger

Focused mode



<https://www.youtube.com/watch?v=IJtUg-3DfUk>

In Summary

- Errors are quite normal for both beginning and advanced programmers. Practice will help, there are no shortcuts!
- First rule: **don't panic!** You'll be able to figure it out.
- Follow the four debugging steps:
 1. Did you spell it correctly?
 2. Is the capitalization/punctuation correct?
 3. Is the syntax correct?
 4. Is the object in your environment?
- Don't forget to **split** the problem.
- **Don't be afraid to ask your peers or instructors for help!!**



Action Items

- 1. Complete Assignments 1.14.**
- 2. Prepare for your first Skill Check!**